

Absolute error



1

a 0.5 kg

b 0.5 g

c \$5

2 $\frac{1}{2}$ mL

3

a 0.05 mL

b 0.5 mm

c 0.05 g

d 0.5 s

e 0.005 kg

f 0.05 m

g 0.005 °C

4 We do not know what place value the number has been rounded to.

The limits of accuracy

5

a 1 cm

b 90 cm

c 0.5 cm

d

i No

ii No

iii No

iv Yes

6

a 10 kg

b 5 kg

c Lower bound: 65 kg, upper bound: 75 kg

7

a 50 km/h

b 5 km/h

c 55 km/h

8

a i 13.455 km

ii 13.445 km

b i 6.45 km

ii 6.35 km

c i 5.5 m

ii 4.5 m

d i 14.005 kg

ii 13.995 kg

e i 4884.5 m

ii 4883.5 m

f i \$52.50

ii \$47.50

9 19.995 m

10 2.05 m

11 a Maximum: 175, minimum: 165



b Maximum: 3050, minimum: 2950

12 a 26.25

b 0.6

13 a 13.85 m/s

b 13.95 m/s

Percentage error

14 a $\pm 0.91\%$ b $\pm 0.96\%$ c $\pm 7.14\%$ d $\pm 1.22\%$
e $\pm 0.29\%$

15 a i 4.7 kg ii 15.5% b i 0.07 m ii 5.3%
c i 7.2 minutes ii 13.8%

16 a 15 891 b 16 109 c 18.9% d 19.2%
e Italy's

Area and volume calculations

17 a 153.9 cm^2 b $132.7 \text{ cm}^2, 176.7 \text{ cm}^2$
c 22.8 cm^2 d 14.8%

18 a Upper bound: 272.6925 m^2 , lower bound: 269.3925 m^2
b Upper bound: 66.2 m, lower bound: 65.8 m

19



- a Lower bound: 9.5 cm, upper bound: 10.5 cm
- b 100 cm^2
- c Lower bound: 90.25 cm^2 , upper bound: 110.25 cm^2
- d 1000 cm^3
- e Lower bound: 857.375 cm^3 , upper bound: 1157.625 cm^3